

PRODUCT REQUIREMENTS DOCUMENT (PRD)

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TABLE OF CONTENTS

- 1. Executive Summary
 - 2. Problem Statement
 - 3. Goals & Objectives
 - 4. Target Audience
 - 5. Scope & Phasing
 - 6. User Personas
 - 7. User Stories
 - 8. Functional Requirements — Student Portal
 - 9. Functional Requirements — Admin Panel
 - 10. Non-Functional Requirements
 - 11. System Architecture
 - 12. Database Schema
 - 13. API Specifications
 - 14. UI/UX Requirements
 - 15. Content Requirements
 - 16. Security Requirements
 - 17. Analytics & Reporting
 - 18. Assumptions & Dependencies
 - 19. Risks & Mitigation
 - 20. Success Metrics (KPIs)
 - 21. Timeline & Milestones
 - 22. Glossary
-

1. EXECUTIVE SUMMARY

MahaBoard Prep is a web-based educational platform designed exclusively for students of the **Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE)**. The platform enables students of **10th Standard (SSC)** and **12th Standard (HSC)** to prepare for their board examinations by:

- Accessing and practicing with **official and model question papers** organized by subject, chapter, and year
- Attempting **objective-type quizzes (MCQ Tests)** chapter-wise and subject-wise
- Tracking their preparation progress through detailed analytics

An **Admin Panel** allows authorized administrators to manage all content: upload question papers, create quizzes, manage users, and monitor platform activity — without requiring any coding knowledge.

2. PROBLEM STATEMENT

Current Situation

Maharashtra Board students (10th and 12th) face significant challenges in exam preparation:

- **Scattered Resources:** Question papers are spread across multiple websites, often unreliable or outdated.
- **No Structured Practice:** No single platform offers chapter-wise MCQ tests aligned to the Maharashtra Board syllabus.
- **Lack of Self-Assessment:** Students cannot gauge their preparation level without proper test environments.
- **Language Barrier:** Most EdTech platforms are built for CBSE students and do not cater to Marathi Medium or Semi-English students.
- **No Progress Tracking:** Students cannot identify weak chapters or subjects efficiently.

The Gap

There is **no dedicated, structured, Maharashtra-Board-specific** exam preparation platform that combines past papers + objective tests + progress analytics in one place.

3. GOALS & OBJECTIVES

Primary Goals

#	Goal	Description
G1	Centralized Resource	Be the go-to platform for all Maharashtra Board 10th & 12th exam prep material
G2	Structured Practice	Provide chapter-wise, subject-wise, and year-wise question paper access
G3	Objective Testing	Enable MCQ quiz practice with instant feedback and detailed results
G4	Admin Autonomy	Allow admins to manage 100% of content without developer help
G5	Student Insights	Help students identify weak areas through performance analytics

Secondary Goals

- Support Marathi and English UI languages
- Achieve 10,000 registered students within 6 months of launch
- Maintain 95%+ platform uptime

4. TARGET AUDIENCE

Primary Users

- **Students:** 10th and 12th Standard students affiliated with Maharashtra State Board
- **Age Range:** 14-18 years
- **Geography:** Maharashtra (Primary), Indian diaspora (Secondary)
- **Language:** English, Marathi, Semi-English (Hindi optional)
- **Device Usage:** 60% Mobile, 30% Laptop/Desktop, 10% Tablet

Secondary Users

- **Admins:** Platform administrators who manage content
- **Teachers/Tutors** (Phase 2): May access to create content for their students

5. SCOPE & PHASING

Phase 1 — MVP (Launch)

- Student registration and login
- Question paper browsing and PDF viewing/download
- Basic MCQ quiz (Practice Mode + Exam Mode)
- Admin panel: Upload papers, create quizzes, manage subjects/chapters
- Student dashboard with basic stats

Phase 2 — Growth (3 months post-launch)

- Leaderboards
- Marathi language UI toggle
- Email notifications
- Advanced analytics and weakness detection
- Scheduled/timed mock tests

Phase 3 — Scale (6 months post-launch)

- Teacher accounts and class management
- Video solution integration
- Discussion forum per chapter
- Mobile App (React Native)
- Offline paper download with DRM

OUT OF SCOPE (v1.0)

- Live tutoring or video classes
 - Payment gateway / subscription model (platform is free at launch)
 - Parent portal
 - CBSE or other board content
-

6. USER PERSONAS

Persona 1 — Priya Sharma (Student)

- **Age:** 17
- **Standard:** 12th (HSC), Science Stream
- **Location:** Pune, Maharashtra

- **Medium:** Semi-English
 - **Device:** Android Smartphone (primary), Laptop (occasional)
 - **Goal:** Score 85%+ in HSC board exams, practice Physics and Chemistry MCQs
 - **Pain Points:** Can't find chapter-wise MCQs; past papers are in different PDFs scattered online
 - **Tech Comfort:** Moderate — uses YouTube, Instagram, WhatsApp daily
 - **Quote:** *"I just want one app where I can give practice tests and know which chapters I'm weak in."*
-

Persona 2 — Rohan Patil (Student)

- **Age:** 15
 - **Standard:** 10th (SSC)
 - **Location:** Nashik, Maharashtra
 - **Medium:** Marathi Medium
 - **Device:** Shared family Android phone
 - **Goal:** Pass SSC with distinction; weak in Mathematics
 - **Pain Points:** English-only interfaces are confusing; limited internet data
 - **Tech Comfort:** Low-Moderate
 - **Quote:** *"Marathi madhe asel tar khup baro hotil." (It would be great if it were in Marathi.)*
-

Persona 3 — Adminji (Platform Admin)

- **Age:** 35
 - **Role:** Educational content manager
 - **Tech Comfort:** Moderate — comfortable with basic web tools, not a developer
 - **Goal:** Upload new papers quickly, create quizzes efficiently, monitor student engagement
 - **Pain Points:** Needs a simple UI; doesn't want to touch code
 - **Quote:** *"I need to upload 20 papers in one session and set a quiz live for tomorrow morning."*
-

7. USER STORIES

Student — Authentication

- As a student, I want to register with my email and password so I can access the platform.
- As a student, I want to set my standard (10th/12th) and medium during onboarding so the platform shows me relevant content.
- As a student, I want to reset my password via email OTP so I can recover my account.

Student — Question Papers

- As a student, I want to browse question papers filtered by standard, subject, chapter, and year so I can find what I need quickly.
- As a student, I want to view a PDF question paper in the browser without downloading so I can start reading immediately.
- As a student, I want to download question papers so I can study offline.
- As a student, I want to bookmark a question paper so I can find it later.
- As a student, I want to mark a paper as "Completed" so I can track what I've already practiced.

Student — Quiz/MCQ Tests

- As a student, I want to take a chapter-wise MCQ quiz in Practice Mode so I get instant feedback after each question.
- As a student, I want to take a full-subject mock test in Exam Mode so I can simulate actual board exam conditions.
- As a student, I want to see a detailed result after each quiz showing my score, wrong answers, correct answers, and explanations.
- As a student, I want to review all questions after the quiz so I can learn from my mistakes.
- As a student, I want to see my quiz history so I can track improvement over time.

Student — Dashboard

- As a student, I want to see my overall progress per subject so I know how much I've covered.
- As a student, I want to see my daily practice streak so I stay motivated.
- As a student, I want to see suggested topics to practice based on my weak areas so I can prioritize my study.

Admin — Content Management

- As an admin, I want to upload a PDF question paper with metadata (subject, chapter, year, etc.) so students can access it.
- As an admin, I want to create a new quiz by adding MCQ questions one by one so students can practice.
- As an admin, I want to import questions from a CSV/Excel file so I can create quizzes faster.
- As an admin, I want to preview a quiz before publishing it so I can verify accuracy.
- As an admin, I want to schedule a quiz to go live at a future date/time so I can plan content in advance.
- As an admin, I want to edit or delete any question paper or quiz so I can correct errors.

Admin — User & Platform Management

- As an admin, I want to view all registered students with their activity so I can monitor engagement.
 - As an admin, I want to block/unblock a student account so I can manage misuse.
 - As an admin, I want to post an announcement visible on student dashboards so I can communicate updates.
 - As an admin, I want to view platform analytics (total users, attempts, top quizzes) so I can make data-driven decisions.
-

8. FUNCTIONAL REQUIREMENTS — STUDENT PORTAL

FR-S1: Authentication & Onboarding

ID	Requirement	Priority
FR-S1.1	Email + password registration	Must Have
FR-S1.2	Email verification via OTP	Must Have
FR-S1.3	Login with JWT session management	Must Have
FR-S1.4	Forgot password / reset via email	Must Have
FR-S1.5	Onboarding: select Standard (10th/12th)	Must Have
FR-S1.6	Onboarding: select Medium (English/Semi-English/Marathi)	Must Have
FR-S1.7	Onboarding: select preferred subjects	Should Have
FR-S1.8	Google OAuth login	Should Have
FR-S1.9	Profile edit (name, standard, medium, avatar)	Must Have

FR-S2: Question Paper Section

ID	Requirement	Priority
FR-S2.1	Browse all available papers	Must Have
FR-S2.2	Filter by: Standard	Must Have
FR-S2.3	Filter by: Subject	Must Have

ID	Requirement	Priority
FR-S2.4	Filter by: Chapter / Topic	Must Have
FR-S2.5	Filter by: Year	Must Have
FR-S2.6	Filter by: Paper Type (Board Exam / Model / Practice)	Should Have
FR-S2.7	Filter by: Medium	Should Have
FR-S2.8	Search papers by keyword	Must Have
FR-S2.9	View PDF inline (embedded PDF viewer)	Must Have
FR-S2.10	Download PDF	Must Have
FR-S2.11	Bookmark a paper	Must Have
FR-S2.12	Mark paper as Completed	Should Have
FR-S2.13	View paper details (marks, duration, year, subject)	Must Have
FR-S2.14	Pagination (20 papers per page)	Must Have
FR-S2.15	Sort by: Latest / Most Attempted / Year	Should Have

FR-S3: Quiz / MCQ Test Section

ID	Requirement	Priority
FR-S3.1	Browse all available quizzes	Must Have
FR-S3.2	Filter quizzes by Standard / Subject / Chapter / Difficulty	Must Have
FR-S3.3	Practice Mode: instant feedback after each question	Must Have
FR-S3.4	Exam Mode: all questions first, results at end	Must Have
FR-S3.5	Countdown timer in Exam Mode	Must Have
FR-S3.6	Auto-submit on timer expiry	Must Have
FR-S3.7	Question navigation panel (jump to any question)	Must Have
FR-S3.8	Mark question for review	Should Have
FR-S3.9	Single-correct MCQ (A/B/C/D options)	Must Have
FR-S3.10	True / False question type	Should Have

ID	Requirement	Priority
FR-S3.11	Image-based questions (question has an image)	Should Have
FR-S3.12	Post-quiz result card: score, percentage, time taken	Must Have
FR-S3.13	Post-quiz: question-wise review (correct/wrong + explanation)	Must Have
FR-S3.14	Post-quiz: subject-wise and chapter-wise breakdown	Should Have
FR-S3.15	Reattempt a quiz	Must Have
FR-S3.16	Save quiz attempt to history	Must Have
FR-S3.17	Share result card (image export)	Nice to Have
FR-S3.18	Leaderboard per quiz (top 10 students)	Nice to Have

FR-S4: Student Dashboard

ID	Requirement	Priority
FR-S4.1	Overview cards: Papers Attempted, Quizzes Taken, Avg. Score, Streak	Must Have
FR-S4.2	Subject-wise progress bar (% completed)	Must Have
FR-S4.3	Recent activity feed	Should Have
FR-S4.4	Bookmarked papers and quizzes quick access	Must Have
FR-S4.5	Daily practice streak counter	Should Have
FR-S4.6	Suggested topics based on weak areas	Should Have
FR-S4.7	Quiz score trend graph (line chart, last 7/30 days)	Should Have
FR-S4.8	Announcements section	Must Have
FR-S4.9	Upcoming scheduled quizzes	Should Have

9. FUNCTIONAL REQUIREMENTS — ADMIN PANEL

FR-A1: Admin Authentication

ID	Requirement	Priority
FR-A1.1	Separate admin login (email + password)	Must Have
FR-A1.2	Role-based access: Super Admin / Content Manager	Should Have
FR-A1.3	Admin session timeout (30 minutes inactivity)	Must Have
FR-A1.4	Admin activity log (who did what, when)	Should Have

FR-A2: Admin Dashboard

ID	Requirement	Priority
FR-A2.1	Total students, papers, quizzes, attempts (live stats)	Must Have
FR-A2.2	Recent uploads and quiz activity	Must Have
FR-A2.3	Top 5 most attempted quizzes	Should Have
FR-A2.4	Top 5 most downloaded papers	Should Have
FR-A2.5	Student growth chart (registrations per day/week/month)	Should Have

FR-A3: Question Paper Management

ID	Requirement	Priority
FR-A3.1	Upload a new question paper (PDF, max 20MB)	Must Have
FR-A3.2	Fill metadata: Standard, Subject, Chapter, Year, Type, Medium, Marks, Duration	Must Have
FR-A3.3	Set status: Draft / Published	Must Have
FR-A3.4	Edit existing paper metadata	Must Have
FR-A3.5	Delete a paper (soft delete)	Must Have

ID	Requirement	Priority
FR-A3.6	View all papers in a searchable, filterable table	Must Have
FR-A3.7	Per-paper analytics: views, downloads, bookmarks	Should Have
FR-A3.8	Bulk upload: CSV metadata + ZIP of PDFs	Nice to Have
FR-A3.9	Duplicate a paper entry	Nice to Have

FR-A4: Quiz / MCQ Management

ID	Requirement	Priority
FR-A4.1	Create a new quiz with metadata (title, standard, subject, chapter, difficulty, time limit, total marks, instructions)	Must Have
FR-A4.2	Add questions to quiz: text, options A/B/C/D, correct answer, explanation, marks	Must Have
FR-A4.3	Add image to a question (upload image)	Should Have
FR-A4.4	Reorder questions via drag-and-drop	Should Have
FR-A4.5	Preview quiz before publishing	Must Have
FR-A4.6	Set quiz status: Draft / Published / Scheduled	Must Have
FR-A4.7	Schedule quiz to go live at a specific date & time	Should Have
FR-A4.8	Edit quiz metadata and questions	Must Have
FR-A4.9	Delete a quiz (soft delete)	Must Have
FR-A4.10	Duplicate quiz to create similar tests	Should Have
FR-A4.11	Bulk import questions from CSV/Excel template	Should Have

ID	Requirement	Priority
FR-A4.12	Export quiz questions to CSV	Nice to Have
FR-A4.13	Per-quiz analytics: attempts, avg. score, per-question correct %	Should Have
FR-A4.14	Enable/disable negative marking	Should Have
FR-A4.15	Set pass percentage for a quiz	Nice to Have

FR-A5: Syllabus / Subject Management

ID	Requirement	Priority
FR-A5.1	Add / Edit / Delete Standards (10th, 12th)	Must Have
FR-A5.2	Add / Edit / Delete Subjects per Standard	Must Have
FR-A5.3	Add / Edit / Delete Chapters per Subject	Must Have
FR-A5.4	Reorder chapters	Should Have
FR-A5.5	View chapter-wise count of papers and quizzes	Should Have

FR-A6: User Management

ID	Requirement	Priority
FR-A6.1	View list of all registered students (name, email, standard, joined date, last active)	Must Have
FR-A6.2	Search and filter students	Must Have
FR-A6.3	View individual student profile with full activity history	Should Have
FR-A6.4	Block / Unblock a student account	Must Have

ID	Requirement	Priority
FR-A6.5	Export user list to CSV	Should Have
FR-A6.6	Delete student account (GDPR compliance)	Should Have

FR-A7: Announcements

ID	Requirement	Priority
FR-A7.1	Create an announcement with title, body, expiry date	Must Have
FR-A7.2	Target announcement: All students / 10th only / 12th only	Should Have
FR-A7.3	Edit and delete announcements	Must Have
FR-A7.4	View list of all announcements with status (Active/Expired)	Must Have

10. NON-FUNCTIONAL REQUIREMENTS

Performance

ID	Requirement	Target
NFR-P1	Page load time (first contentful paint)	< 2.5 seconds
NFR-P2	API response time (95th percentile)	< 500ms
NFR-P3	PDF load time (inline viewer)	< 4 seconds
NFR-P4	Quiz submission processing time	< 1 second
NFR-P5	Support concurrent users	500 simultaneous

Availability & Reliability

ID	Requirement	Target
NFR-A1	Platform uptime	99.5% monthly
NFR-A2	Data backup frequency	Daily automated backups

ID	Requirement	Target
NFR-A3	Backup retention	30 days
NFR-A4	Recovery Time Objective (RTO)	< 4 hours

Scalability

- System should handle 10,000 registered users without architectural changes
- Horizontal scaling must be possible via containerization (Docker)
- CDN integration for static assets and PDF delivery

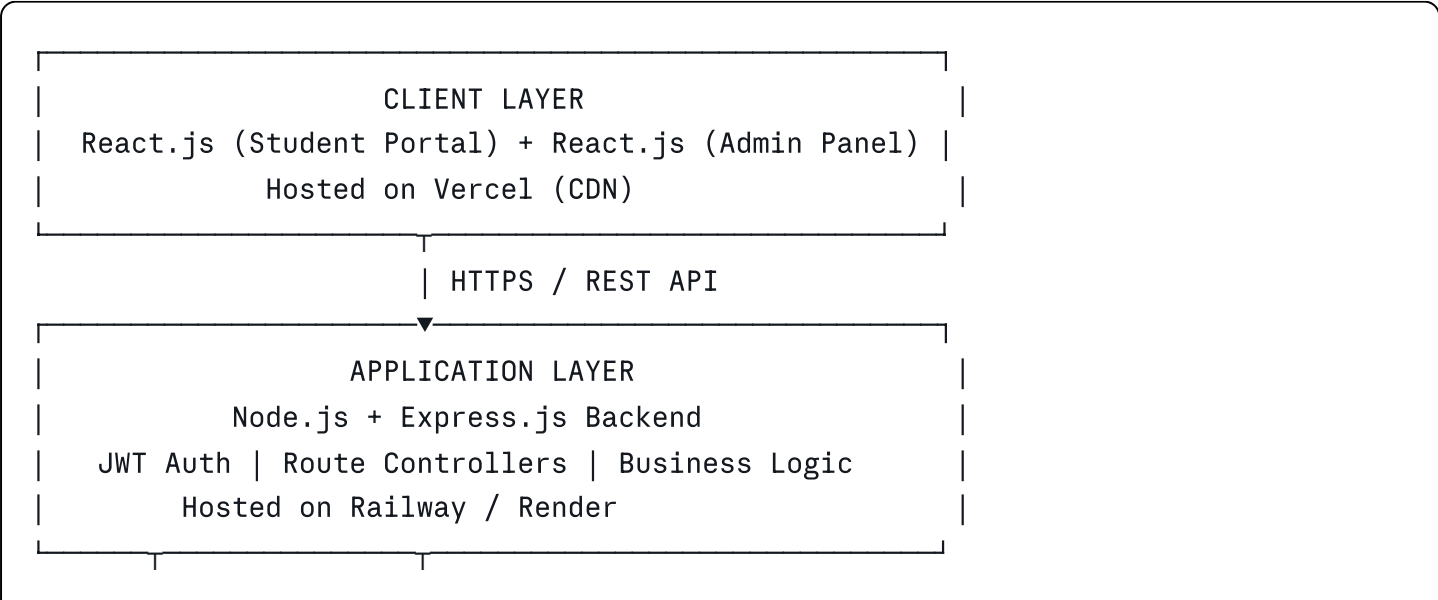
Compatibility

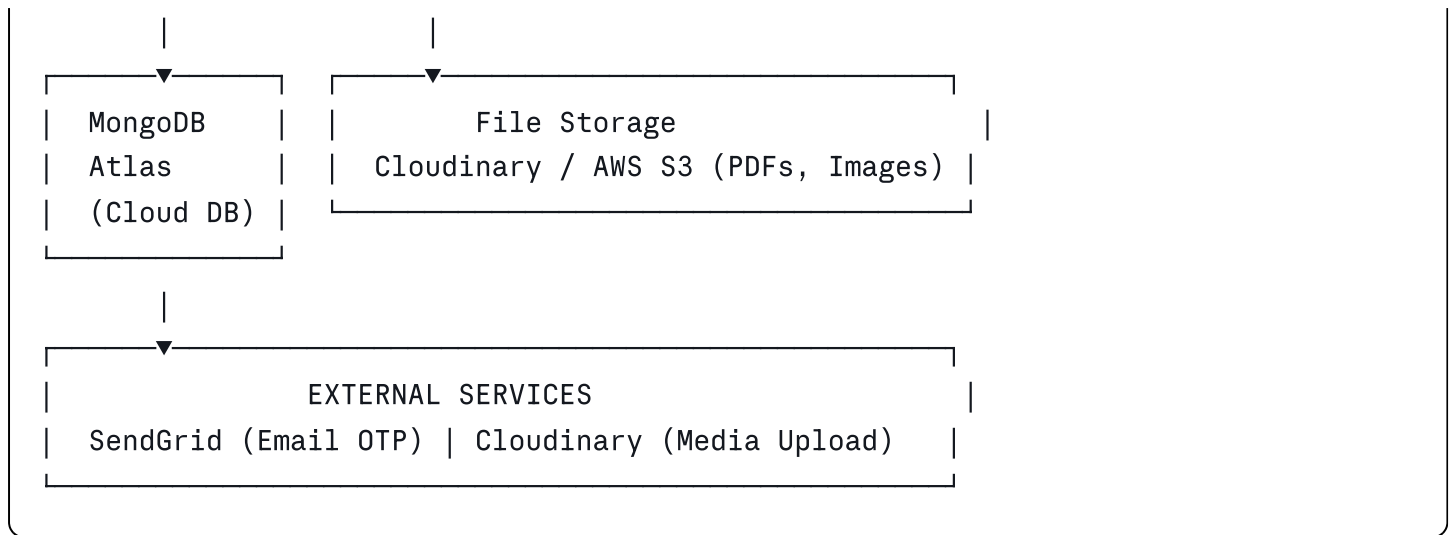
Platform	Requirement
Browsers	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Mobile OS	Android 8.0+, iOS 13+
Screen Sizes	320px (mobile) to 2560px (4K desktop)

Accessibility

- WCAG 2.1 Level AA compliance
- Keyboard navigable interface
- Screen reader compatible (ARIA labels)
- Minimum contrast ratio 4.5:1 for text

11. SYSTEM ARCHITECTURE





12. DATABASE SCHEMA

Collection: users

```
{
  _id: ObjectId,
  name: String (required),
  email: String (unique, required),
  password: String (hashed),
  role: Enum ["student", "admin"],
  standard: Enum ["10", "12"],
  medium: Enum ["english", "semi-english", "marathi"],
  subjects: [String],
  isVerified: Boolean,
  isBlocked: Boolean,
  streak: Number,
  lastActiveAt: Date,
  createdAt: Date
}
```

Collection: standards

```
{
  _id: ObjectId,
  name: String,           // "10th Standard (SSC)"
  code: Enum ["10", "12"],
  order: Number
}
```

Collection: subjects

```
{
  _id: ObjectId,
  name: String,          // "Mathematics"
  standardId: ObjectId,
  medium: String,
  icon: String,          // emoji or icon key
  color: String,          // hex color for UI
  order: Number,
  isActive: Boolean
}
```

Collection: chapters

```
{
  _id: ObjectId,
  name: String,          // "Chapter 3 - Arithmetic Progression"
  number: Number,
  subjectId: ObjectId,
  order: Number,
  isActive: Boolean
}
```

Collection: papers

```
{
  _id: ObjectId,
  title: String,
  standardId: ObjectId,
  subjectId: ObjectId,
  chapterId: ObjectId,
  year: Number,
  type: Enum ["board", "model", "practice"],
  medium: String,
  pdfUrl: String,        // Cloudinary/S3 URL
  pdfPublicId: String,   // Cloudinary ID for deletion
  marks: Number,
  durationMinutes: Number,
  status: Enum ["draft", "published"],
  uploadedBy: ObjectId, // admin user ID
  analytics: {
    views: Number,
    downloads: Number,
    bookmarks: Number
  }
}
```



```
    },
    createdAt: Date,
    updatedAt: Date
  }
}
```

Collection: quizzes

```
{
  _id: ObjectId,
  title: String,
  standardId: ObjectId,
  subjectId: ObjectId,
  chapterId: ObjectId,
  difficulty: Enum ["easy", "medium", "hard"],
  timeLimitMinutes: Number,
  totalMarks: Number,
  negativeMarking: Boolean,
  negativeMarkValue: Number,
  passPercentage: Number,
  instructions: String,
  status: Enum ["draft", "published", "scheduled"],
  scheduledAt: Date,
  createdBy: ObjectId,
  analytics: {
    totalAttempts: Number,
    avgScore: Number,
    avgTimeTaken: Number
  },
  createdAt: Date,
  updatedAt: Date
}
```

Collection: questions

```
{
  _id: ObjectId,
  quizId: ObjectId,
  text: String,
  imageUrl: String,      // optional
  type: Enum ["mcq", "true_false"],
  optionA: String,
  optionB: String,
  optionC: String,
  optionD: String,
  correctOption: Enum ["A", "B", "C", "D"],
}
```

```
  explanation: String, // optional
  marks: Number,
  order: Number,
  analytics: {
    totalAttempts: Number,
    correctAttempts: Number
  }
}
```

Collection: attempts

```
{
  _id: ObjectId,
  userId: ObjectId,
  quizId: ObjectId,
  mode: Enum ["practice", "exam"],
  answers: [
    {
      questionId: ObjectId,
      selectedOption: Enum ["A", "B", "C", "D", null],
      isCorrect: Boolean,
      marksAwarded: Number,
      timeTakenSeconds: Number
    }
  ],
  totalScore: Number,
  maxScore: Number,
  percentage: Number,
  timeTakenSeconds: Number,
  isCompleted: Boolean,
  submittedAt: Date,
  startedAt: Date
}
```

Collection: bookmarks

```
{
  _id: ObjectId,
  userId: ObjectId,
  type: Enum ["paper", "quiz"],
  refId: ObjectId,
  createdAt: Date
}
```

Collection: announcements

```
{
  _id: ObjectId,
  title: String,
  body: String,
  targetAudience: Enum ["all", "10", "12"],
  expiresAt: Date,
  isActive: Boolean,
  createdBy: ObjectId,
  createdAt: Date
}
```

13. API SPECIFICATIONS

Base URL: `https://api.mahaboardprep.in/v1`

Authentication Endpoints

Method	Endpoint	Description
POST	/auth/register	Student registration
POST	/auth/login	Student/Admin login
POST	/auth/verify-email	Email OTP verification
POST	/auth/forgot-password	Send reset OTP
POST	/auth/reset-password	Reset password with OTP
GET	/auth/me	Get current user profile
PUT	/auth/profile	Update user profile
POST	/auth/logout	Invalidate token

Papers Endpoints

Method	Endpoint	Description
GET	/papers	List papers (with filters & pagination)
GET	/papers/:id	Get single paper details

Method	Endpoint	Description
POST	/papers	Upload new paper (Admin only)
PUT	/papers/:id	Update paper metadata (Admin only)
DELETE	/papers/:id	Delete paper (Admin only)
GET	/papers/:id/download	Increment download count + return URL

Quiz Endpoints

Method	Endpoint	Description
GET	/quizzes	List quizzes (with filters)
GET	/quizzes/:id	Get quiz details + questions
POST	/quizzes	Create quiz (Admin only)
PUT	/quizzes/:id	Update quiz (Admin only)
DELETE	/quizzes/:id	Delete quiz (Admin only)
POST	/quizzes/:id/attempt	Start a quiz attempt
POST	/quizzes/:id/submit	Submit quiz attempt
GET	/quizzes/:id/attempts	Get all attempts for a quiz (Admin)

User & Progress Endpoints

Method	Endpoint	Description
GET	/users/dashboard	Student dashboard data
GET	/users/attempts	Student's quiz attempt history
GET	/users/bookmarks	Student's bookmarks
POST	/users/bookmarks	Add bookmark
DELETE	/users/bookmarks/:id	Remove bookmark
GET	/admin/users	List all students (Admin)
GET	/admin/users/:id	View student detail (Admin)
PUT	/admin/users/:id/block	Block/unblock student (Admin)

Syllabus Endpoints

Method	Endpoint	Description
GET	/subjects	List subjects (filter by standard)
GET	/subjects/:id/chapters	List chapters for a subject
POST	/admin/subjects	Create subject (Admin)
POST	/admin/chapters	Create chapter (Admin)

Announcement Endpoints

Method	Endpoint	Description
GET	/announcements	Get active announcements
POST	/admin/announcements	Create announcement (Admin)
PUT	/admin/announcements/:id	Update announcement (Admin)
DELETE	/admin/announcements/:id	Delete announcement (Admin)

14. UI/UX REQUIREMENTS

Design System

Color Palette:

Token	Color	Hex	Usage
Primary	Deep Blue	 #1E3A8A	Buttons, headers, nav
Accent	Saffron Orange	 #F97316	CTA, highlights, badges
Success	Green	 #16A34A	Correct answers, pass status
Error	Red	 #DC2626	Wrong answers, errors
Neutral	Gray-100 to Gray-900	—	Text, backgrounds, borders
Background	Off-white	 #F8FAFC	Page background

Typography:

- Headings: Poppins (Google Font)
- Body: Inter (Google Font)
- Minimum body size: 14px
- Line height: 1.6

Spacing: 8px base grid system

Border Radius: 8px (cards), 4px (inputs), 999px (pills/badges)

Key Screens

Student Portal:

1. Landing / Home Page (unauthenticated)
2. Register / Login Page
3. Onboarding Flow (Standard + Medium + Subjects)
4. Student Dashboard
5. Question Papers List (Browse + Filter)
6. Question Paper Detail + PDF Viewer
7. Quiz List (Browse + Filter)
8. Quiz Detail Page (instructions + start button)
9. Quiz Attempt Page (Practice Mode)
10. Quiz Attempt Page (Exam Mode with timer)
11. Quiz Result Page
12. Quiz Review Page (question-by-question)
13. Bookmarks Page
14. Profile & Settings Page

Admin Panel:

1. Admin Login Page
2. Admin Dashboard
3. Papers List (table view)
4. Upload Paper Form
5. Edit Paper Form
6. Quizzes List (table view)
7. Create Quiz Form (with question builder)
8. Edit Quiz Form
9. Quiz Preview Page

- 10. Subject & Chapter Management
 - 11. Students List (table)
 - 12. Student Detail Page
 - 13. Announcements Management
 - 14. Platform Analytics Page
-

Responsive Breakpoints

Breakpoint	Width	Target
Mobile	< 640px	Android/iPhone phones
Tablet	640px - 1024px	iPad, large phones
Desktop	> 1024px	Laptops, desktops

15. CONTENT REQUIREMENTS

10th Standard (SSC) — Subjects

- 1. Mathematics (Algebra + Geometry)
- 2. Science & Technology Part 1 (Physics + Chemistry)
- 3. Science & Technology Part 2 (Biology + Environmental)
- 4. English (First Language)
- 5. Marathi (Second Language)
- 6. Hindi (Third Language)
- 7. History & Political Science
- 8. Geography
- 9. Sanskrit (optional)

12th Standard (HSC) — Science Stream

- 1. Physics
- 2. Chemistry
- 3. Biology
- 4. Mathematics & Statistics

12th Standard (HSC) — Commerce Stream

- 1. Economics

2. Accountancy & Financial Management
3. Organisation of Commerce & Management
4. Mathematics & Statistics

12th Standard (HSC) — Arts Stream

1. History
2. Geography
3. Political Science
4. Sociology
5. Economics
6. Psychology

Common Subjects (All Streams — HSC)

1. English (Compulsory)
2. Marathi (Compulsory)
3. Hindi (Optional)

Minimum Content at Launch (Seed Data)

- 5 past papers per subject per standard = ~65 papers
- 3 quizzes per subject (10 MCQs each) = ~60 quizzes
- All subjects and chapters pre-configured in admin panel
- 2 sample announcements

16. SECURITY REQUIREMENTS

ID	Requirement
SEC-1	All API endpoints served over HTTPS (TLS 1.2+)
SEC-2	Passwords hashed using bcrypt (salt rounds ≥ 12)
SEC-3	JWT access token expiry: 15 minutes; refresh token: 7 days
SEC-4	Role-based middleware on every protected route
SEC-5	Admin routes require additional X-Admin-Secret header
SEC-6	File upload validation: PDF only, max 20MB, virus scan
SEC-7	Rate limiting: 100 requests/minute per IP

ID	Requirement
SEC-8	Input sanitization (prevent XSS and NoSQL injection)
SEC-9	CORS whitelist to allowed frontend domains only
SEC-10	No sensitive data (passwords, tokens) in logs
SEC-11	MongoDB Atlas IP whitelist (only backend server IP)
SEC-12	Environment variables for all secrets (never hardcoded)
SEC-13	Student cannot access admin API routes (403 Forbidden)
SEC-14	GDPR-compliant: student data deletion on request

17. ANALYTICS & REPORTING

Student-Facing Analytics

- Overall quiz attempts and average score
- Subject-wise performance breakdown
- Chapter-wise score history
- Quiz score trend over time (graph)
- Weakest chapters (lowest average score)
- Practice streak

Admin Analytics

- Total registrations (daily / weekly / monthly trend)
- Daily active users
- Total quiz attempts per quiz
- Per-question correct attempt rate (identify bad/ambiguous questions)
- Top 10 most attempted quizzes
- Top 10 most downloaded papers
- Subject-wise content engagement
- Student retention (7-day and 30-day return rate)

Reports Available to Admin

- Student list export (CSV)
- Quiz attempts report (per quiz, date range)
- Content inventory report (all papers and quizzes with metadata)

18. ASSUMPTIONS & DEPENDENCIES

Assumptions

- All question papers are in PDF format and copyright-cleared for use by the platform
- Admin has access to internet and a modern browser for the Admin Panel
- Students primarily access the platform via mobile (60%+)
- Content will be uploaded by admin manually at launch (bulk upload in Phase 2)
- Platform is free for students at v1.0 launch (no payment gateway needed)
- Maharashtra Board syllabus is stable and won't change mid-development

Dependencies

Dependency	Purpose	Provider
MongoDB Atlas	Cloud database	MongoDB Inc.
Cloudinary	PDF and image storage	Cloudinary
SendGrid	Email OTP and notifications	Twilio SendGrid
Vercel	Frontend hosting	Vercel Inc.
Railway / Render	Backend hosting	Railway / Render
Google Fonts	Typography (Poppins, Inter)	Google
React PDF Viewer	In-browser PDF display	Open source

19. RISKS & MITIGATION

#	Risk	Likelihood	Impact	Mitigation
R1	Copyright issues with uploaded question papers	Medium	High	Only upload papers from official MSBSHSE sources; add copyright disclaimer
R2	Low student adoption	Medium	High	Launch with referral program; partner with coaching classes in Pune/Mumbai/Nashik
R3	Admin panel abuse (wrong content uploaded)	Low	High	Content review workflow; staging before publish

#	Risk	Likelihood	Impact	Mitigation
R4	PDF storage costs exceed budget	Low	Medium	Implement file size limits; use Cloudinary free tier + upgrade plan
R5	Server downtime during exam season	Medium	High	Auto-scaling + CDN caching for PDFs; load testing pre-season
R6	Data loss	Low	Critical	Daily MongoDB Atlas backups; point-in-time recovery enabled
R7	Mobile performance issues	Medium	Medium	Mobile-first design; lazy loading; paginate all lists
R8	Quiz cheating / sharing answers	Low	Low	Randomize question order; randomize option order per attempt

20. SUCCESS METRICS (KPIs)

Growth Metrics

Metric	1 Month	3 Months	6 Months
Registered Students	500	3,000	10,000
Daily Active Users	100	600	2,500
Papers Available	50	150	400
Quizzes Available	60	200	500

Engagement Metrics

Metric	Target
Avg. session duration	> 12 minutes
Quiz completion rate	> 75%
7-day user retention	> 40%
Daily returning users	> 30% of DAU
Paper download rate	> 50% of views

Quality Metrics

Metric	Target
Page load time	< 2.5 seconds
Bug reports per week	< 5 after Month 1
Quiz result accuracy	100% (zero wrong corrections)
Uptime	> 99.5%

21. TIMELINE & MILESTONES

Phase	Milestone	Duration	Completion Target
Planning	PRD finalized, tech stack confirmed	Week 1-2	Week 2
Design	Wireframes + UI mockups (all screens)	Week 3-4	Week 4
Backend	Database models + all API routes	Week 5-7	Week 7
Auth	JWT authentication system (student + admin)	Week 6	Week 7
Frontend	Student portal — all pages	Week 7-11	Week 11
Frontend	Admin panel — all pages	Week 9-12	Week 12
Integration	Frontend-Backend integration + file upload	Week 12-13	Week 13
Content	Seed data: all subjects, chapters, papers, quizzes	Week 13-14	Week 14
Testing	QA testing (functional, performance, security)	Week 14-15	Week 15
Bug Fix	Critical bug resolution	Week 15-16	Week 16
Deploy	Production deployment + monitoring setup	Week 16	Week 16
Launch	Public Beta Launch	—	Week 16
Post-Launch	Phase 2 planning + user feedback iteration	Week 17+	Ongoing

Total Development Timeline: ~16 Weeks (4 Months)

22. GLOSSARY

Term	Definition
SSC	Secondary School Certificate — Maharashtra Board 10th Standard exam
HSC	Higher Secondary Certificate — Maharashtra Board 12th Standard exam
MSBSHSE	Maharashtra State Board of Secondary and Higher Secondary Education
MCQ	Multiple Choice Question — a question with 4 options, one correct answer
Practice Mode	Quiz mode where feedback is given after each question immediately
Exam Mode	Quiz mode where all questions are shown first; result shown at end
Admin Panel	Secured back-office interface for content and user management
Attempt	One complete session of a student taking a quiz
Bookmark	Student's saved reference to a paper or quiz for later access
Chapter-wise	Organized or filtered specifically by textbook chapter
Draft	Content created but not yet visible to students
Published	Content that is live and visible to all students
Scheduled	Content set to automatically publish at a future date/time
Soft Delete	Marking content as deleted without removing it from the database
Streak	Number of consecutive days a student has practiced on the platform
JWT	JSON Web Token — a secure method for transmitting authentication data
CDN	Content Delivery Network — distributes content globally for fast load times
DRM	Digital Rights Management — controls access to digital content

Document End

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